

Geometric AI Architecture: Toward a Modular, Spline-Based Intelligence Framework

Abstract

This paper introduces *Geometric AI Architecture* (GAA), a framework that reconceptualizes artificial intelligence memory and skill representation through the mathematical formalism of Non-Uniform Rational B-Splines (NURBS), B-Splines, and Bézier curves. Rather than treating intelligence as a monolithic weight matrix embedded within a large language model (LLM), GAA encodes discrete cognitive units—memories, domain expertise, and reasoning heuristics—as parametric spline segments, each defined by a compact triplet of control points, weights, and knot vectors. This approach draws direct inspiration from recent advances in spline-parameterized neural networks, most prominently Kolmogorov-Arnold Networks (KANs), in which every weight parameter is replaced by a learnable univariate function parametrized as a B-spline, enabling faster neural scaling laws and superior interpretability compared to multi-layer perceptrons (MLPs).¹ Unlike prior geometric neural frameworks that remain theoretical, GAA has been implemented as a proprietary, production-grade system and empirically validated against operational data from infrastructure finance and renewable energy deployment workflows, with findings drawn from the Mountain Shifts Library of 15 Case Studies.

Central to the framework is the concept of *memory slicing*: the selective extraction of high-density semantic significance from completed project cycles, encoded as NURBS control-point manifolds. This mirrors established work on differentiable NURBS modules, which demonstrate that control points, knot vectors, and weights can be analytically differentiated, integrated into deep learning pipelines, and used to enforce geometric constraints at orders of magnitude improved efficiency over conventional surface representations.² GAA encodes project-specific lessons—capital expenditure patterns, foreign exchange shocks, construction phase dynamics—as exportable `.nurbs` segments that can be re-imported into new project contexts without model retraining. The result is a *Library of Curves*: a Memory Vault in which each knowledge artifact maintains its own resolution fidelity, allowing lossless export for internal use, lossy B-Spline approximations for partner sharing, and Bézier trend extractions for investor-grade reporting.

The *Skill-as-a-Segment* sub-framework addresses on-demand domain intelligence. Professional expertise—such as Asset-Backed Finance structures, UL/ASTM certification workflows, or royalty agreement logic—is represented as pre-computed NURBS patches that compose with a base B-Spline reasoning graph at near-zero latency. This is conceptually aligned with recent physics-informed deep B-Spline networks, which demonstrate that B-spline control-point networks are universal approximators of arbitrary-dimensional partial differential equations, supporting the claim that spline-parameterized modules can serve as closed, portable reasoning units.³ The architecture also draws upon the theoretical foundation of SpliNet, which decouples

network parameterization from numerical integration schemes by embedding B-Spline basis functions directly into neural network layers—a structural analogy for the decoupling of skill knowledge from base model weights in GAA.⁴

Memory persistence and retrieval are handled through a Graph-Retrieval Augmented Generation (GraphRAG) layer, wherein indexed spline segments function as structured nodes within a dynamic knowledge graph. This is consistent with emerging evidence that graph-based retrieval substantially mitigates LLM hallucination, enhances domain-specific factual accuracy, and enables context-aware generation beyond what dense vector retrieval alone can provide.⁵ The retrieval pipeline maps directly onto GraphRAG's Graph-Based Indexing (G-Indexing) and Graph-Guided Retrieval (G-Retrieval) stages,⁶ operating on the geometric structure of the memory segments rather than on dense vector embeddings.

A distinguishing contribution of GAA is its architecture of *dimensional ontological projection*. By licensing a curated suite of domain ontologies—spanning Health, U.S. and California Law, S&P market data (2000–2025), U.S. Supreme Court jurisprudence (1950–2024), Finance (Tiers I and II), Agriculture, FDA regulatory science, and computational systems—the system converts the Memory Vault from a flat retrieval index into a multi-dimensional semantic reasoning space. Each ontology defines an independent coordinate axis along which any spline memory segment can be simultaneously projected, enabling the same encoded knowledge to be queried through a legal lens, a financial lens, and a regulatory lens in parallel. This produces composite cross-domain analysis, projections, and reports that single-domain retrieval systems cannot generate. The investment in ontological coverage is therefore not a data licensing cost but a geometric infrastructure cost: each additional ontology adds a dimension to the reasoning manifold.

The paper also introduces *Geometric Pruning*, a resolution-tiered export protocol that allows practitioners to tune the fidelity of a memory segment to its intended audience. This concept extends analogous multi-resolution work in NeuroNURBS, which achieves an 86.7% reduction in GPU consumption and a 79.9% drop in memory requirements by encoding 3D solid geometry directly as NURBS parameters rather than UV-grid point samples.⁷ The same principle of parametric compression is applied here to semantic memory, where a full project cycle is compressed into a navigable spline manifold. Transformer-based approaches to B-Spline knot placement,⁸ as well as hybrid B-Spline/neural-network operator architectures that leverage the B-Spline convex hull property for real-time uncertainty quantification,⁹ further establish the technical basis for the Geometric Adapter layer.

Finally, the paper situates GAA within a broader paradigm of collective machine intelligence. The *Collective Ownership Memory* protocol enables organizations to build versioned knowledge vaults segmented by project type, macroeconomic cycle, and fund strategy, each encoded in its native spline format and empirically grounded in the Mountain Shifts Library of 15 Case Studies. This advances a vision of AI infrastructure in which intelligence is not merely stored but *geometrically structured*, enabling modular composition, transparent auditability, and zero-

latency domain specialization. Taken together, GAA offers a rigorous geometric foundation for the next generation of interpretable, portable, and composable artificial intelligence systems.

Keywords: Geometric AI Architecture, NURBS, B-Spline, Kolmogorov-Arnold Networks, modular memory, ontological projection, GraphRAG, Memory Vault, Collective Ownership Memory, Geometric Pruning

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